CLOSED-FORM GAUSSIAN INFORMATION PROJECTION FOR EMBEDDED HYPERCUBES AND BETA–GAUSSIAN MIXTURES

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Abstract. We derive closed-form Gaussian information projections of Beta–Gaussian distributions whose tangential components are supported on affine embedded hypercubes and whose normal components are Gaussian. The optimal Gaussian is uniquely characterized by moment matching, and the corresponding minimum Kullback–Leibler divergence admits an explicit expression depending only on the intrinsic Beta parameters, independently of the embedding geometry, ambient dimension, and normal Gaussian parameters. In the uniform case, this divergence reduces to a linear function of the intrinsic dimension. We further extend this construction to finite, pure embedded cubical complexes by considering mixtures of the componentwise Gaussian projections. Using the joint convexity of the Kullback–Leibler divergence, we establish an explicit upper bound on the approximation error of the resulting Gaussian Mixture. For uniform tangential distributions and volume-based mixture weights, the bound is linear in the intrinsic dimension and independent of the geometry of the complex, its embedding, and the normal Gaussian parameters.

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Contents

1. Introduction 1
2. Construction 2
3. Conclusion 10
4. Acknowledgment 10
References 10
Authors' Affiliations 11

1. Introduction

Gaussian *Probability Density Function*s (PDFs) provide a tractable representation of probability measures, but their approximation quality can deteriorate when the underlying PDFs exhibit bounded or geometrically constrained tangential structure. This motivates the study of Gaussian information projections for PDFs associated with embedded geometric structures.

In this work, we consider affine embeddings of the unit hypercube into an ambient Euclidean space. We equip the reference hypercube with product-Beta PDFs and introduce independent Gaussian PDFs in the orthogonal complement of the embedding's tangent space.

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This construction gives absolutely continuous PDFs with respect to ambient Lebesgue measure, while preserving the prescribed Beta PDF structure in the tangential directions. We refer to these PDFs as Beta–Gaussian PDFs.

Our main result provides a closed-form characterization of the unique Gaussian information projection of a Beta–Gaussian PDFs through moment matching. Moreover, the resulting minimum *Kullback–Leibler* (KL) divergence admits an explicit expression depending only on the intrinsic Beta PDF parameters, independently of the embedding geometry, ambient dimension, and normal Gaussian parameters. In the uniform case, the divergence reduces to a linear function of the intrinsic dimension.

We further extend this construction to finite, pure embedded cubical complexes by considering mixtures of the componentwise Beta–Gaussian PDFs and their corresponding Gaussian information projections. Using the joint convexity of the KL divergence, we establish an explicit upper bound for the approximation error of the resulting *Gaussian Mixture* (GM). In the uniform case, with volume-based mixture weights, this bound is again linear in the intrinsic dimension and independent of the geometry of the complex, its embedding, and the normal Gaussian PDF parameters.

2. Construction

Definition 2.1 (Convex Set [1])**.** *Let $d \in \mathbb{N}$. A set $K \subseteq \mathbb{R}^d$ is called convex if, for every $x, y \in K$ and every $t \in [0,1]$, the convex combination*

$$(1-t)x + ty \tag{2.1}$$

belongs to K.

Definition 2.2 (Standard Cube [2])**.** *For $d \in \mathbb{N}$, the standard d-dimensional cube is the set*

$$Q_d := [0,1]^d = \{x \in \mathbb{R}^d : 0 \le x_i \le 1 \text{ for } i = 1, \dots, d\}. \tag{2.2}$$

Definition 2.3 (Embedded Cube)**.** *Let $1 \le d \le D$, let $v \in \mathbb{R}^D$, and let $E \in \mathbb{R}^{D \times d}$ have full column rank, that is,*

$$\operatorname{rank}(E) = d. \tag{2.3}$$

The affine map

$$\Phi : \mathbb{R}^d \to \mathbb{R}^D, \quad \Phi(x) := v + Ex, \tag{2.4}$$

is injective. The image of the standard d-dimensional cube under Φ,

$$C := \Phi(Q_d) = \{v + Ex : x \in Q_d\}, \tag{2.5}$$

is called an embedded d-dimensional cube in $\mathbb{R}^D$.

Definition 2.4 (Tangent and Normal Spaces)**.** *Let $C = \Phi(Q_d) \subseteq \mathbb{R}^D$ be an embedded d-dimensional cube, with*

$$\Phi(x) = v + Ex, \quad v \in \mathbb{R}^D, \quad E \in \mathbb{R}^{D \times d}, \quad \operatorname{rank}(E) = d. \tag{2.6}$$

We equip $\mathbb{R}^D$ with its standard Euclidean inner product

$$\langle x, y \rangle := x^\top y, \quad x, y \in \mathbb{R}^D. \tag{2.7}$$

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The tangent space of C, understood as the linear direction space of its affine hull, is defined by

$$T_C \coloneqq \operatorname{range}(E) = \{Eh : h \in \mathbb{R}^d\} \subseteq \mathbb{R}^D. \tag{2.8}$$

Equivalently, since Φ is affine, its differential is constant and satisfies

$$D\Phi(x) = E \qquad \text{for every } x \in \mathbb{R}^d, \tag{2.9}$$

so that

$$T_C = \operatorname{range}(D\Phi(x)) \quad \text{for every } x \in \mathbb{R}^d. \tag{2.10}$$

The normal space of C, relative to the ambient Euclidean space $\mathbb{R}^D$, is defined as the orthogonal complement of its tangent space:

$$N_C \coloneqq T_C^{\perp} = \left\{ z \in \mathbb{R}^D : \langle z, w \rangle = 0 \text{ for every } w \in T_C \right\}. \tag{2.11}$$

Equivalently, the normal space is the null space of the transpose of E:

$$N_C = \ker(E^{\top}) = \left\{ z \in \mathbb{R}^D : E^{\top} z = 0 \right\}. \tag{2.12}$$

In particular, the ambient space admits the orthogonal direct sum decomposition

$$\mathbb{R}^D = T_C \oplus N_C, \quad T_C \perp N_C, \tag{2.13}$$

with dimensions

$$\dim(T_C) = d, \quad \dim(N_C) = D - d. \tag{2.14}$$

Remark 2.5 (Geometric Interpretation)**.** *The embedded cube C is a d-dimensional parallelotope contained in the affine subspace*

$$\operatorname{aff}(C) = v + \operatorname{range}(E) = v + T_C. \tag{2.15}$$

Its d edge vectors are precisely the columns of E. In particular, the columns of E are linearly independent, but they are not necessarily mutually orthogonal or of equal length. Consequently, the term embedded cube refers to the affine image of the standard cube and does not imply that C is an orthogonal cube in the ambient Euclidean metric. The translation vector v determines its position, whereas E determines its orientation, edge directions, and edge lengths.

Proposition 2.6. *Let $C = F_C([0,1]^d)$, where $F_C(x) = v + Dx$ and $D \in \mathbb{R}^{n \times d}$ has full column rank. Then*

$$\operatorname{vol}_d(C) = \sqrt{\det(D^{\top} D)}. \tag{2.16}$$

Proof. By the area formula (see [3, Theorem 3.2.3, p. 243]),

$$\operatorname{vol}_d(C) = \int_{[0,1]^d} \sqrt{\det(D^{\top} D)}\, dx = \sqrt{\det(D^{\top} D)}. \tag{2.17}$$

□

Definition 2.7 (Faces of an Affine Cube)**.** *Let $C = v + E[0,1]^d$ be an embedded affine d-cube. For $I \subseteq \{1, \ldots, d\}$ and $\epsilon \in \{0,1\}^I$, define*

$$F_{I,\epsilon}(C) \coloneqq v + E\left\{ x \in [0,1]^d : x_i = \epsilon_i \text{ for all } i \in I \right\}. \tag{2.18}$$

These sets are the faces of C, with

$$\dim F_{I,\epsilon}(C) = d - |I|. \tag{2.19}$$

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Definition 2.8 (Embedded Cubical Complex)**.** *A finite collection $\mathcal{K}$ of embedded affine cubes is an embedded cubical complex if*

(1) $F \in \mathcal{K}$ whenever F is a face of $C \in \mathcal{K}$;
(2) $C \cap C'$ is either empty or a face of both $C, C' \in \mathcal{K}$.

The complex is pure of dimension d if

$$\dim C = d \quad \text{for every inclusion-maximal } C \in \mathcal{K}. \tag{2.20}$$

Its realization and maximal cells are denoted by

$$|\mathcal{K}| \coloneqq \bigcup_{C \in \mathcal{K}} C, \quad \mathcal{K}^{[d]} \coloneqq \{C \in \mathcal{K} : \ C \text{ is inclusion-maximal}\}. \tag{2.21}$$

Proposition 2.9 (Cubical Volume Decomposition)**.** *Let $\mathcal{K}$ be a finite, pure, d-dimensional embedded cubical complex in $\mathbb{R}^D$. Then*

$$\mathcal{H}^d(|\mathcal{K}|) = \sum_{C \in \mathcal{K}^{[d]}} \mathrm{vol}_d(C). \tag{2.22}$$

Moreover, if $0 < \mathcal{H}^d(|\mathcal{K}|) < \infty$, then

$$\mathcal{U}_{|\mathcal{K}|} = \sum_{C \in \mathcal{K}^{[d]}} \frac{\mathrm{vol}_d(C)}{\mathcal{H}^d(|\mathcal{K}|)} \mathcal{U}_C, \quad \mathcal{U}_A \coloneqq \frac{\mathcal{H}^d|_A}{\mathrm{vol}_d(A)}. \tag{2.23}$$

Proof. Distinct maximal cells intersect in proper faces, which are $\mathcal{H}^d$-null by [4, Lemma 2.2, p. 92]. Finite additivity of $\mathcal{H}^d$ then follows from [3, Theorem 2.1.3, p. 54]. □

Definition 2.10 (Product-Beta PDF [5])**.** *For parameter vectors $a = (a_1, \dots, a_d) \in (0, \infty)^d$ and $b = (b_1, \dots, b_d) \in (0, \infty)^d$, let $X = (X_1, \dots, X_d)^\top$ have independent coordinates*

$$X_i \sim \mathrm{Beta}(a_i, b_i), \quad i = 1, \dots, d. \tag{2.24}$$

Its PDF with respect to Lebesgue measure on $[0, 1]^d$ is

$$f_{a,b}(x) = \prod_{i=1}^{d} \frac{x_i^{a_i - 1}(1 - x_i)^{b_i - 1}}{B(a_i, b_i)}, \quad x \in (0, 1)^d, \tag{2.25}$$

where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is the beta function [6, § 5.12, p. 142]. The PDF is set to zero outside $[0, 1]^d$.

Proposition 2.11 (Moments and Entropy of a Product-Beta Law)**.** *Under definition 2.10, define, for each i,*

$$\mu_i = \frac{a_i}{a_i + b_i}, \quad s_i^2 = \frac{a_i b_i}{(a_i + b_i)^2(a_i + b_i + 1)}. \tag{2.26}$$

Then

$$\mathbb{E}(X) = \mu, \quad \mathbb{V}(X) = V \coloneqq \mathrm{diag}(s_1^2, \dots, s_d^2) \succ 0. \tag{2.27}$$

The differential entropy is finite and equals

$$h(X) = \sum_{i=1}^{d} h_i, \tag{2.28}$$

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where

$$h_i = \log B(a_i, b_i) - (a_i - 1)\psi(a_i) - (b_i - 1)\psi(b_i) + (a_i + b_i - 2)\psi(a_i + b_i),$$

and ψ is the digamma function.

Proof. The moment identities are the standard univariate Beta moment formulas, together with independence. For entropy, taking the expectation of the logarithm of the PDF gives

$$-h(X_i) = \mathbb{E}(\log f_{X_i}(X_i)) = -\log B(a_i, b_i) + (a_i - 1)\mathbb{E}[\log X_i] + (b_i - 1)\mathbb{E}(\log(1 - X_i)). \tag{2.29}$$

Differentiating $\log B(a, b)$ with respect to its two arguments, or equivalently differentiating the beta integral under the integral sign, gives

$$\mathbb{E}(\log X_i) = \psi(a_i) - \psi(a_i + b_i), \quad \mathbb{E}(\log(1 - X_i)) = \psi(b_i) - \psi(a_i + b_i). \tag{2.30}$$

Substitution gives eq. (2.28). These expectations are finite for all strictly positive parameters: near either endpoint, $x^{a_i - 1}|\log x|$ and $(1 - x)^{b_i - 1}|\log(1 - x)|$ are integrable. Thus each marginal entropy is finite. Independence gives additivity of entropy and eq. (2.28). □

Definition 2.12 (Beta Law on an Embedded Hypercube)**.** *Let $C = v + E[0, 1]^d$ be an embedded hypercube. For $a, b \in (0, \infty)^d$, the embedded product-Beta law on C is the pushforward of the product-Beta law in definition 2.10 under $F_C(x) = v + Ex$. In other words,*

$$Y_T = v + EX, \qquad X_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i). \tag{2.31}$$

We denote its law by $P_T(C; a, b)$. Its intrinsic PDF is understood with respect to $\mathcal{H}^d$ on the affine hull of C.

Proposition 2.13 (Embedded Moments and Intrinsic Entropy)**.** *Under definition 2.12,*

$$m_T \coloneqq \mathbb{E}(Y_T) = v + E\mu, \quad \Sigma_T \coloneqq \mathbb{V}(Y_T) = EVE^\top. \tag{2.32}$$

The covariance is positive definite on T_C and zero on N_C. The intrinsic differential entropy is

$$h_{T_C}(Y_T) = \sum_{i=1}^{d} h_i + \frac{1}{2}\log\det(E^\top E). \tag{2.33}$$

Proof. The moment formulas follow from the affine transformation rule. Since E has full column rank and $V \succ 0$, the restriction of $EVE^\top$ to $T_C = \text{range}(E)$ is positive definite. For the entropy, the constant d-dimensional Jacobian of F_C is $J_C = \sqrt{\det(E^\top E)}$. The change-of-variables formula on the affine hull gives $h_{T_C}(Y_T) = h(X) + \log J_C$, which is eq. (2.33). □

Definition 2.14 (Beta–Gaussian PDF)**.** *Let $C = v + E[0, 1]^d \subseteq \mathbb{R}^D$ be an embedded hypercube, with $d \leq D$. Choose matrices $U_T \in \mathbb{R}^{D \times d}$ and $U_N \in \mathbb{R}^{D \times (D-d)}$ whose columns are orthonormal bases of T_C and N_C, respectively. Let*

$$X_i \overset{\text{ind}}{\sim} \text{Beta}(a_i, b_i), \quad Z_N \sim \mathcal{N}_{D-d}(\nu_N, S_N), \qquad S_N \succ 0, \tag{2.34}$$

with X and Z_N independent. Define

$$W \coloneqq v + EX + U_N Z_N. \tag{2.35}$$

The law of W is called a Beta–Gaussian distribution associated with C, and is denoted by $P(C; a, b, \nu_N, S_N)$. If $d = D$, the normal term is absent and all statements involving U_N, Z_N, S_N are interpreted on the zero-dimensional space.

Lemma 2.15 (Absolute Continuity and Finite Entropy)**.** *The Beta–Gaussian* Random Variable *(RV) in definition 2.14 is absolutely continuous with respect to D-dimensional Lebesgue measure and has finite differential entropy. More precisely, for $d < D$,*

$$h(W) = \sum_{i=1}^{d} h_i + \frac{1}{2}\log\det(E^\top E) + \frac{D-d}{2}\log(2\pi e) + \frac{1}{2}\log\det S_N. \tag{2.36}$$

Proof. The product-Beta vector has a PDF on $(0,1)^d$ and finite entropy by proposition 2.11. Its affine image has finite intrinsic entropy by proposition 2.13. The Gaussian normal coordinate has a nondegenerate PDF and finite entropy. Under the orthogonal affine coordinate map

$$\Phi(w) = \begin{pmatrix} U_T^\top (w - v) \\ U_N^\top (w - v) \end{pmatrix}, \tag{2.37}$$

the absolute Jacobian determinant is one and

$$\Phi(W) = \begin{pmatrix} U_T^\top EX \\ Z_N \end{pmatrix}. \tag{2.38}$$

The two blocks are independent. Their product law is absolutely continuous in $\mathbb{R}^D$, and entropy additivity gives eq. (2.36). The same argument without the normal block applies when $d = D$. □

Remark 2.16. *The normal Gaussian is essential when $d < D$: without it the distribution is supported on a d-dimensional affine subspace and is singular with respect to ambient Lebesgue measure. The positive-definite normal covariance makes the ambient density nondegenerate. No isotropy assumption is imposed on S_N.*

Definition 2.17 (Forward KL divergence)**.** *For Lebesgue PDFs p, q on $\mathbb{R}^D$, define*

$$\mathrm{KL}(P : Q) = \int_{\mathbb{R}^D} p(x)\log\frac{p(x)}{q(x)}\mathrm{d}x, \tag{2.39}$$

with the usual extended-real convention. Let $\mathcal{G}_D$ denote the family of nondegenerate Gaussian PDFs on $\mathbb{R}^D$. The information projection of P onto $\mathcal{G}_D$ is any minimizer of $\mathrm{KL}(P : Q)$ over $Q \in \mathcal{G}_D$.

Proposition 2.18 (Gaussian Information Projection by Moment Matching)**.** *Let P be absolutely continuous on $\mathbb{R}^D$, have finite differential entropy, finite second moments, and positive-definite covariance Σ_P. Then its unique information projection onto $\mathcal{G}_D$ is*

$$Q^* = \mathcal{N}_D(\cdot \mid m_P, \Sigma_P), \qquad m_P = \mathbb{E}_P[X]. \tag{2.40}$$

Proof. For $Q = \mathcal{N}_D(\cdot \mid m, \Sigma)$, direct integration gives

$$\mathrm{KL}(P : Q) = -h(P) + \frac{D}{2}\log(2\pi) + \frac{1}{2}\log\det\Sigma + \frac{1}{2}\mathrm{tr}(\Sigma^{-1}\Sigma_P) + \frac{1}{2}(m - m_P)^\top\Sigma^{-1}(m - m_P).$$

For fixed $\Sigma \succ 0$, the last term is uniquely minimized at $m = m_P$. To minimize over Σ, set $A = \Sigma_P^{-1/2}\Sigma\Sigma_P^{-1/2}$. Up to constants independent of A, the objective is $\log\det A + \mathrm{tr}(A^{-1})$. If $\lambda_1, \ldots, \lambda_D$ are the positive eigenvalues of A, this is $\sum_k(\log\lambda_k + \lambda_k^{-1})$, whose unique minimum occurs at $\lambda_k = 1$ for every k (the scalar function has derivative $(\lambda - 1)/\lambda^2$ and a unique global minimum at one). Thus $\Sigma = \Sigma_P$, proving existence and uniqueness. □

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Theorem 2.19 (Closed-form Projection of a Beta–Gaussian Law)**.** *Let $P = P(C; a, b, \nu_N, S_N)$ be the Beta–Gaussian law in definition 2.14. Define μ, V by eq.* (2.26)*, and let $\overline{\Sigma}_N = U_N S_N U_N^\top$ (zero if $d = D$). Then the unique Gaussian information projection of P onto $\mathcal{G}_D$ is*

$$Q^* = \mathcal{N}_D(\cdot \mid m, \Sigma), \tag{2.41}$$

where

$$m = v + E\mu + U_N \nu_N, \quad \Sigma = EVE^\top + \overline{\Sigma}_N. \tag{2.42}$$

Its minimum divergence is

$$\mathrm{KL}(P : Q^*) = C_d(a, b) := \sum_{i=1}^{d} \left[\frac{1}{2} \log(2\pi e s_i^2) - h_i \right], \tag{2.43}$$

where s_i^2 and h_i are given by eqs. (2.26) *and* (2.28)*. In particular, the minimum divergence is independent of the ambient dimension, the embedding v, E, the hypercube volume, the normal mean, and the normal covariance.*

Proof. Independence of X and Z_N gives the mean and covariance in eq. (2.42). The tangential covariance is positive definite on T_C, while $\overline{\Sigma}_N$ is positive definite on N_C; since these spaces are orthogonal complements, $\Sigma \succ 0$. The entropy is finite by lemma 2.15, and the second moments are finite because the Beta component is bounded and the normal component is Gaussian. Hence proposition 2.18 applies and proves the stated unique projection.

It remains to compute the divergence. In orthonormal coordinates adapted to $T_C \oplus N_C$, both P and Q^* factor into a tangential and a normal distribution. Their normal factors agree exactly, so invariance and additivity of KL divergence give

$$\mathrm{KL}(P : Q^*) = \mathrm{KL}(P_T : Q_T^*). \tag{2.44}$$

The entropy of the tangential target is, by eq. (2.33),

$$h_{T_C}(Y_T) = \sum_{i=1}^{d} h_i + \frac{1}{2} \log\det(E^\top E). \tag{2.45}$$

Let U_T be an orthonormal basis matrix for T_C and put $R = U_T^\top E$. Since $E = U_T R$ and R is invertible, the matrix of the tangential covariance in this orthonormal basis is $RVR^\top$. Therefore

$$\det_{T_C}(\Sigma_T) = \det(R)^2 \det(V) = \det(E^\top E) \prod_{i=1}^{d} s_i^2.$$

The entropy of the moment-matched tangential Gaussian is consequently

$$h(Q_T^*) = \frac{d}{2} \log(2\pi e) + \frac{1}{2} \log\det(E^\top E) + \frac{1}{2} \sum_{i=1}^{d} \log s_i^2. \tag{2.46}$$

Subtracting the target entropy and using eq. (2.44) gives eq. (2.43). All geometric and normal terms cancel, as claimed. □

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Corollary 2.20 (Uniform PDF on an Embedded Hypercube)**.** *If $a_i = b_i = 1$ for all i, then X is uniform on $[0,1]^d$, $\mu_i = 1/2$, $s_i^2 = 1/12$, and $h_i = 0$. Hence*

$$m = v + \frac{1}{2}E\mathbf{1}_d + U_N\nu_N, \quad \Sigma = \frac{1}{12}EE^\top + \overline{\Sigma}_N, \tag{2.47}$$

and

$$\mathrm{KL}(P : Q^*) = \frac{d}{2}\log\left(\frac{\pi e}{6}\right). \tag{2.48}$$

The divergence depends only on the intrinsic dimension d.

Proof. For the uniform Beta$(1,1)$ law, the stated moments and entropy follow immediately from the univariate formulas. Substitution into theorem 2.19 gives the result. □

Remark 2.21 (Nonnegativity)**.** *Each summand in eq.* (2.43) *is the KL divergence between a univariate Beta(a_i, b_i) density and its moment-matched Gaussian on $\mathbb{R}$. It is therefore nonnegative. Equality cannot occur for a nondegenerate Beta law, since its support is bounded whereas a nondegenerate Gaussian has unbounded support. Thus $C_d(a,b) > 0$ for $d \geq 1$.*

Definition 2.22 (Finite GM)**.** *A finite GM on $\mathbb{R}^D$ is a PDF*

$$q(x) = \sum_{j=1}^{M} \lambda_j \mathcal{N}_D(x \mid m_j, \Sigma_j), \quad \lambda_j \geq 0, \quad \sum_{j=1}^{M} \lambda_j = 1, \quad \Sigma_j \succ 0. \tag{2.49}$$

Proposition 2.23 (Joint Convexity of KL divergence [7, Proposition 2.14, p. 92])**.** *Let p_j, q_j be PDFs with respect to a common measure and let $\lambda_j \geq 0$ with $\sum_j \lambda_j = 1$. Then*

$$\mathrm{KL}\left(\sum_{j=1}^{M} \lambda_j p_j : \sum_{j=1}^{M} \lambda_j q_j\right) \leq \sum_{j=1}^{M} \lambda_j \,\mathrm{KL}(p_j : q_j), \tag{2.50}$$

where the inequality is understood in the extended-real sense.

Proof. For almost every x, apply the log-sum inequality [7, Lemma 2.10, p. 93]

$$\left(\sum_j r_j\right) \log \frac{\sum_j r_j}{\sum_j s_j} \leq \sum_j r_j \log \frac{r_j}{s_j} \tag{2.51}$$

with $r_j = \lambda_j p_j(x)$ and $s_j = \lambda_j q_j(x)$, using the standard zero conventions. Integration gives eq. (2.50). □

Theorem 2.24 (Beta–Gaussian Mixture Approximation on an Embedded Cubical Complex)**.** *Let $\mathcal{K}$ be a finite, pure, d-dimensional embedded cubical complex in $\mathbb{R}^D$, with maximal cells $C_1, \ldots, C_M$. For each j, write*

$$C_j = v_j + E_j[0,1]^d, \tag{2.52}$$

where E_j has full column rank. Choose parameters $a_j, b_j \in (0,\infty)^d$, a normal mean $\nu_{N,j}$, and a positive-definite normal covariance $S_{N,j}$ on N_{C_j} (with the normal term omitted if $d = D$). Let P_j be the corresponding Beta–Gaussian law and let q_j be its unique Gaussian information projection as specified by theorem 2.19. For arbitrary mixture weights

$$\lambda_j > 0, \qquad \sum_{j=1}^{M} \lambda_j = 1, \tag{2.53}$$

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define

$$P = \sum_{j=1}^{M} \lambda_j P_j, \quad Q = \sum_{j=1}^{M} \lambda_j q_j. \tag{2.54}$$

Then each q_j is the unique Gaussian information projection of P_j, and

$$\mathrm{KL}(P : Q) \le \sum_{j=1}^{M} \lambda_j C_d(a_j, b_j), \tag{2.55}$$

where C_d is given in eq. (2.43)*. In particular, the bound is independent of the ambient dimension, the embeddings of the cells, and all normal means and covariance matrices.*

Proof. The componentwise information-projection statement and the identity $\mathrm{KL}(P_j : q_j) = C_d(a_j, b_j)$ follow from theorem 2.19. Applying joint convexity, proposition 2.23, to the mixtures in eq. (2.54) gives

$$\mathrm{KL}(P : Q) \le \sum_{j=1}^{M} \lambda_j \,\mathrm{KL}(P_j : q_j) = \sum_{j=1}^{M} \lambda_j C_d(a_j, b_j), \tag{2.56}$$

which proves the bound. Its stated invariances follow term by term from theorem 2.19. □

Corollary 2.25 (Uniform Cubical–Complex Approximation)**.** *Under the assumptions of theorem 2.24, suppose $0 < \mathcal{H}^d(|\mathcal{K}|) < \infty$. On each maximal cell let the tangential law be uniform, i.e. $a_{j,i} = b_{j,i} = 1$ for all i, j, and choose the volume weights*

$$\lambda_j = \frac{\mathrm{vol}_d(C_j)}{\mathcal{H}^d(|\mathcal{K}|)}. \tag{2.57}$$

Let $P = \sum_j \lambda_j P_j$ be the resulting mixture after independent normal Gaussian smoothing on each cell, and let $Q = \sum_j \lambda_j q_j$ be the mixture of the componentwise Gaussian information projections. Then

$$\mathrm{KL}(P : Q) \le \frac{d}{2} \log\left(\frac{\pi e}{6}\right). \tag{2.58}$$

The bound depends only on the intrinsic dimension d and is independent of the ambient dimension, the geometry and embedding of the cubical complex, and the normal Gaussian parameters.

Proof. By proposition 2.9, the volume weights are positive and sum to one. By corollary 2.20, every component divergence equals $\frac{d}{2}\log(\pi e/6)$. Substitution into theorem 2.24 proves eq. (2.58). □

Remark 2.26 (What the Mixture Bound Does and Does not Assert)**.** *The mixture in theorem 2.24 is obtained by projecting each component separately and retaining the original mixture weights. The theorem proves a constructive upper bound for this particular GM. It does not assert that Q globally minimizes $\mathrm{KL}(P : \widetilde{Q})$ over all GMs $\widetilde{Q}$, nor that the displayed upper bound is the optimal value of such a global mixture optimization problem.*

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Remark 2.27 (Uniform Measure Versus its Smoothed Version)**.** *The volume-weighted identity is an identity of probability measures with respect to d-dimensional Hausdorff measure on the cubical complex. If $d < D$, this unsmoothed measure is singular with respect to D-dimensional Lebesgue measure. The KL guarantee in corollary 2.25 concerns the mixture after adding the componentwise normal Gaussian perturbations, which produces ambient Lebesgue densities; it is not a finite ambient-density KL statement for the original singular uniform measure.*

3. Conclusion

We established a closed-form characterization of the forward Gaussian information projection of Beta–Gaussian PDFs associated with affine embedded hypercubes. The optimal Gaussian PDF is uniquely determined by moment matching, and the corresponding minimum KL divergence admits an explicit expression depending only on the intrinsic Beta parameters, independently of the embedding geometry, ambient dimension, and normal Gaussian PDF parameters. In the uniform PDF case, this divergence reduces to a linear function of the intrinsic dimension.

We further extended this construction to finite, pure embedded cubical complexes by constructing GMs from componentwise information projections. Joint convexity of the KL divergence gives an explicit upper bound on the representation divergence of the resulting GM. For uniform tangential PDFs and volume-based mixture weights, the bound reduces to a linear function of the intrinsic dimension, independently of the geometry and embedding of the cubical complex and the normal Gaussian PDF parameters.

These results provide a dimension-independent framework for quantifying Gaussian PDF and GM representation divergence of PDFs associated with embedded geometric structures.

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Symbol Legend
⊛: Idea, □: Math, #: Programming, ¶: Writing, ⊥: Critique

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